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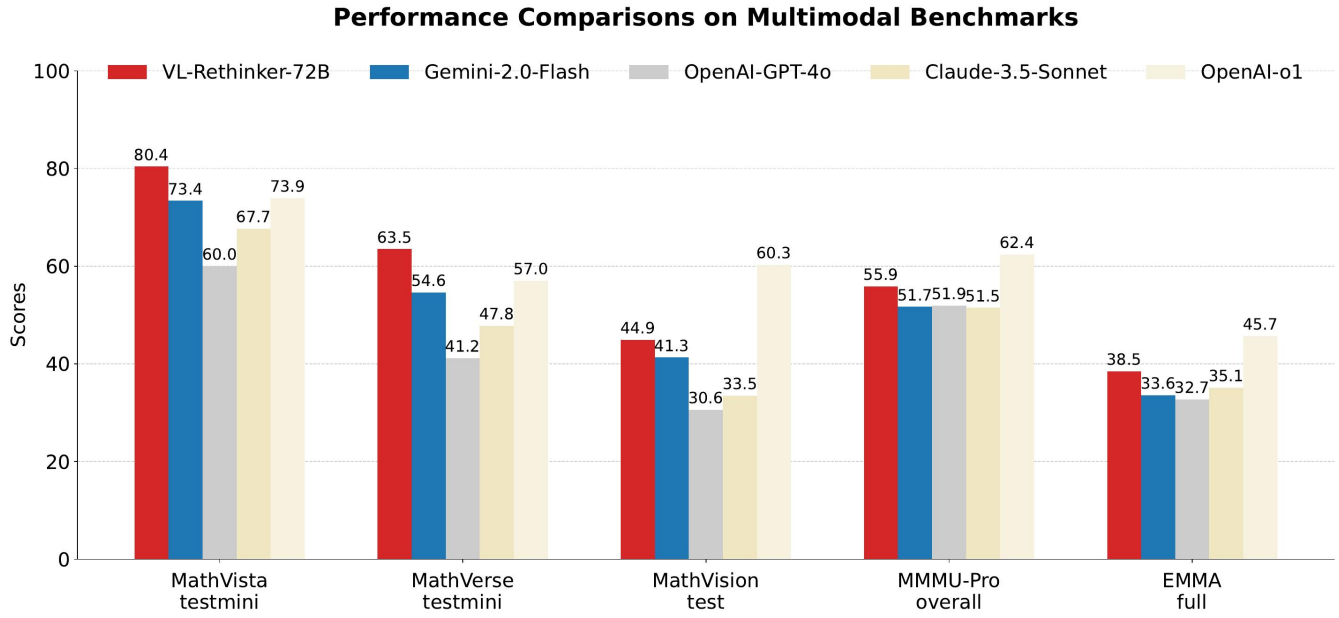


Figure 1: **VL-Rethinker-72B compared with proprietary models.** Accuracy (%) on MathVista testmini, MathVerse testmini, MathVision test, MMMU-Pro overall and EMMA full (values as in Table 1). VL-RETHINKER-72B scores higher than GPT-4o, Claude-3.5-Sonnet and Gemini-2.0-Flash on all five benchmarks and higher than OpenAI-o1 on MathVista and MathVerse; o1 remains ahead on MathVision, MMMU-Pro and EMMA.

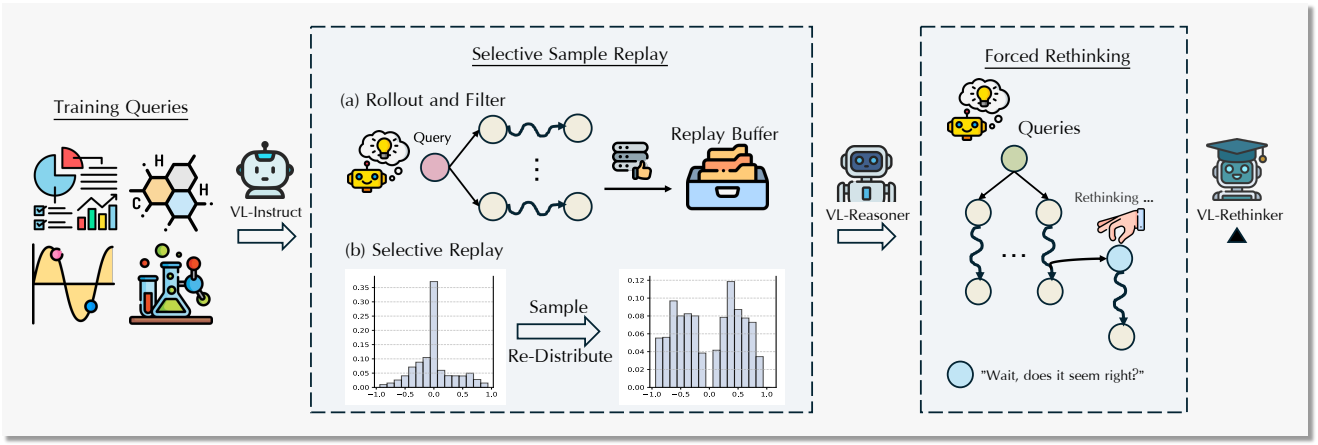


Figure 2: **Method overview.** Starting from an instruction-tuned VLM (VL-Instruct), RL with Selective Sample Replay (SSR) produces VL-Reasoner: (a) rollouts for each query are scored with a binary correctness reward and samples with non-zero advantage are kept in a replay buffer; (b) replay samples from the buffer with priority increasing in |advantage|, redistributing the training batch away from the spike at zero advantage (left histogram) towards informative samples (right histogram). Forced Rethinking then appends a rethinking trigger (e.g., "Wait, does it seem right?") to a fraction of rollouts during RL, yielding VL-RETHINKER.

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
Proprietary Models							
OpenAI-o1	73.9	57.0	60.3	62.4	78.2	45.7	56.2
OpenAI-GPT-4o	60.0	41.2	30.6	51.9	69.1	32.7	52.7
Claude-3.5-Sonnet	67.7	47.8	33.5	51.5	68.3	35.1	52.3
Gemini-2.0-Flash	73.4	54.6	41.3	51.7	70.7	33.6	54.1
Open-Source Models							
Llama4-Scout-109B	70.7	-	-	52.2	69.4	24.6	31.8
InternVL-2.5-78B	72.3	51.7	34.9	48.6	61.8	27.1	44.1
QvQ-72B	71.4	48.6	35.9	51.5	**70.3**	32.0	8.8
LLaVA-OV-72B	67.5	39.1	30.1	31.0	56.8	23.8	29.7
Qwen-2.5-VL-32B	74.7	48.5	38.4	49.5	†59.4	31.1	13.3
Qwen-2.5-VL-72B	74.8	57.2	38.1	51.6	†67.0	34.1	49.0
VL-Rethinker-32B	78.8	56.9	40.5	50.6	65.6	37.9	19.9
VL-Rethinker-72B	**80.4**	**63.5**	**44.9**	**55.9**	68.8	**38.5**	**51.3**
Δ (VL-Rethinker-72B − best other open)	+5.6	+6.3	+6.5	+3.7	−1.5	+4.4	+2.3

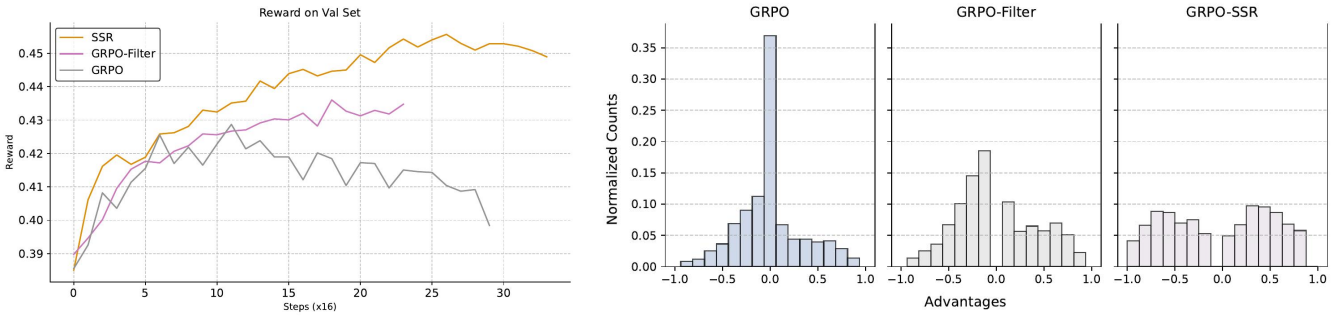
Table 1: **Comparison of VL-Rethinker-72B with proprietary and open-source models.** Accuracy (%) on each benchmark split. Within the open-source block, **bold** marks the best score and underline the best score of another open-source model, the reference for the Δ row (on MMMU, QvQ-72B is both). Proprietary models are listed for reference and are not marked; OpenAI-o1 remains ahead on MathVision, MMMU-Pro, MMMU, EMMA and MEGA. †: reproduced with our evaluation protocol; "-": not reported.

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
General Vision-Language Models							
InternVL2-8B	58.3	-	17.4	29.0	51.2	19.8	26.0
InternVL2.5-8B	64.4	39.5	19.7	34.3	56.0	-	30.4
Qwen2-VL-7B	58.2	-	16.3	30.5	54.1	20.2	34.8
Qwen2.5-VL-7B	68.2	46.3	25.1	36.9	†54.3	21.5	35.0
LLaVA-OV-7B	63.2	26.2	-	24.1	48.8	18.3	22.9
Kimi-VL-16B	68.7	44.9	21.4	-	†55.7	-	-
Vision-Language Reasoning Models							
MM-Eureka-8B (InternVL)	67.1	40.4	22.2	27.8	49.2	-	-
MM-Eureka-7B (Qwen)	73.0	50.3	26.9	-	-	-	-
R1-VL-7B	63.5	40.0	24.7	7.8	44.5	8.3	29.9
R1-Onevision-7B	64.1	46.4	29.9	21.6	-	20.8	27.1
OpenVLThinker-7B	70.2	47.9	25.3	37.3	52.5	26.6	12.0
VL-Rethinker-7B	**74.9**	**54.2**	**32.3**	**41.7**	**56.7**	**29.7**	**37.2**
Δ (VL-Rethinker-7B − best other)	+1.9	+3.9	+2.4	+4.4	+0.7	+3.1	+2.2

Table 2: **Comparison of VL-Rethinker-7B with general and reasoning vision-language models of similar size.** Accuracy (%) on each benchmark split. **Bold**: best score; underline: best score of any other listed model, the reference for the Δ row. †: reproduced with our evaluation protocol; "-": not reported.

Model	RL algorithm	Forced Rethinking	Training data	MathVision	MathVista	MathVerse	MMMU-Pro	EMMA
VL-RETHINKER-7B	GRPO-SSR	✓	16K	**32.3**	**74.9**	**54.2**	**41.7**	**29.7**
VL-Reasoner-7B (w/o Forced Rethinking)	GRPO-SSR	✗	16K	29.8	72.4	53.2	40.9	29.5
– SSR (GRPO-Filter)	GRPO-Filter	✗	16K	28.5	72.0	50.0	40.0	26.9
– SSR – Filter (vanilla GRPO)	GRPO	✗	16K	26.0	70.9	51.4	38.8	26.2
– text-only data	GRPO-SSR	✗	13K	29.1	73.5	53.5	41.1	28.7
– science and text-only data	GRPO-SSR	✗	11K	28.0	71.6	50.3	39.7	28.0

Table 3: **Ablation of Forced Rethinking, Selective Sample Replay (SSR) and the training-data mix** at 7B scale. Accuracy (%) on MathVision test, MathVista testmini, MathVerse testmini, MMMU-Pro overall and EMMA full. Indented rows each remove the named component(s) from VL-Reasoner-7B, the SSR-trained model before Forced Rethinking. GRPO-Filter discards zero-advantage samples and refills the batch uniformly from the current step's non-zero-advantage samples, without SSR's cross-step buffer and |advantage|-prioritised replay. **Bold**: best per column.



(a) Validation reward during training.

(b) Distribution of advantages in training batches.

Figure 3: **Training dynamics of GRPO, GRPO-Filter and GRPO-SSR.** (a) Mean reward (accuracy) on the held-out validation set; the x-axis counts logged points, one every 16 training steps. Vanilla GRPO peaks at about 0.43 near point 11 (≈180 steps), then declines to about 0.40 by point 29. GRPO-Filter keeps improving to about 0.435 over its shorter run (23 points, ≈370 steps). GRPO-SSR improves fastest and reaches about 0.455 around points 23–26 (≈370–420 steps). Each curve is a single run. (b) Normalised histograms of the advantages of the samples used for policy updates. Under GRPO, about 37% of samples sit in the bin at zero advantage. Filtering removes exact zeros but leaves the mass concentrated at small |advantage|. SSR's |advantage|-prioritised replay shifts the mass towards |advantage| ≈ 0.3–0.9.

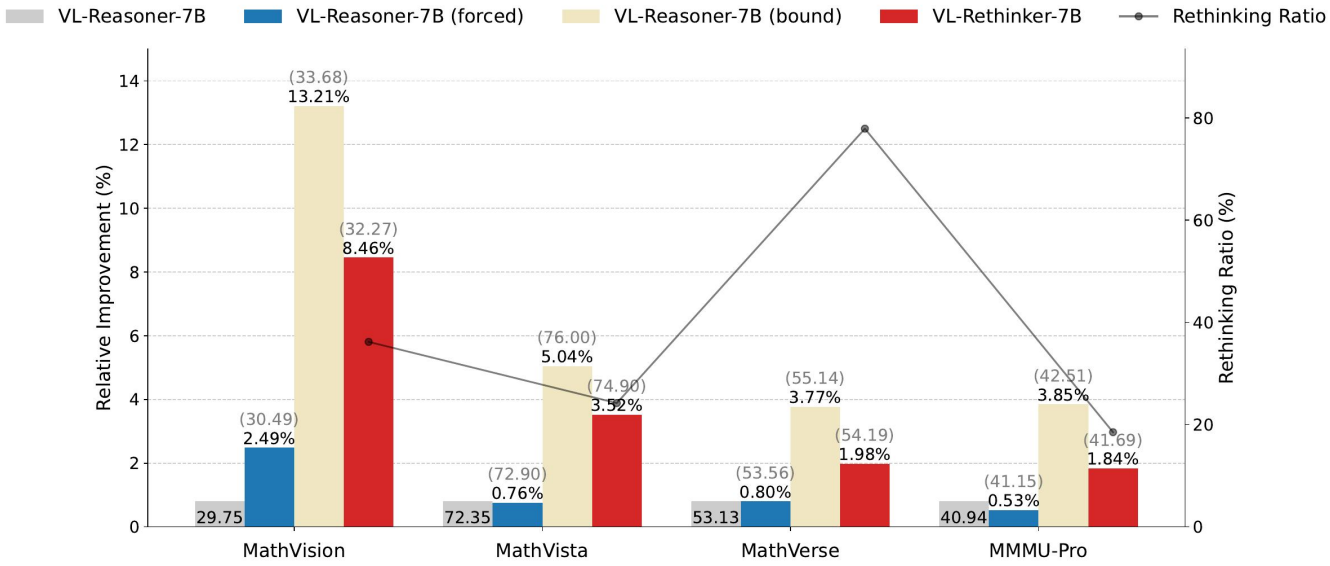


Figure 4: **Relative improvement with different rethinking strategies at 7B.** Bars (left axis) give the relative improvement (%) over VL-Reasoner-7B, the SSR-trained model without Forced Rethinking (gray, its accuracy printed at the bar base). Absolute accuracies are in parentheses. Blue: VL-Reasoner-7B forced to rethink at test time. Beige: an oracle upper bound of test-time rethinking for VL-Reasoner-7B, in which the rethought answer replaces the original only when it is correct. Red: VL-Rethinker-7B, trained with Forced Rethinking. The line (right axis) gives the fraction of VL-Rethinker-7B responses that contain a self-reflection phrase (e.g., "wait"), read from the plot. Test-time forcing alone gives small gains (0.5–2.5%), whereas training with Forced Rethinking gives 1.8–8.5%, with the largest gain on MathVision. VL-Rethinker-7B rethinks selectively rather than uniformly: about 78% of responses on MathVerse, 36% on MathVision, and about 24% and 18% on MathVista and MMMU-Pro.



Figure 5: **Vanishing advantages in GRPO training.** Fraction of training queries per step (x-axis: training steps $\times 16$) whose rollouts are all correct (green), all incorrect (gray) or mixed ("effective queries", purple; dashed: linear trend). Only mixed groups receive a non-zero group-normalised advantage. Over training the effective fraction falls from about 0.41 to 0.20, while the all-correct fraction rises from about 0.19 to 0.47.

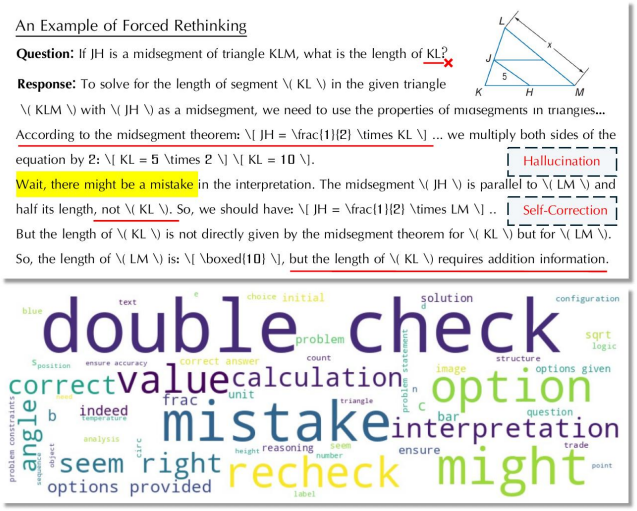


Figure 6: **Forced Rethinking in practice.** *Top:* a response from training with Forced Rethinking. The initial reasoning misapplies the midsegment theorem (JH $= \frac{1}{2}$KL, a hallucinated relation). After the appended trigger "Wait, there might be a mistake", the model notices that JH is parallel to and half of LM, not KL, and revises its conclusion (self-correction). *Bottom:* word cloud of the rethinking segments, dominated by verification and correction phrases ("double check", "mistake", "recheck", "might", "seem right").